A Survey of Hallucination Evaluation for Large Language Models: Taxonomy, Reliability, and Open Challenges

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Abstract

Large language models can produce fluent but unsupported, unverifiable, or factually incorrect content. Although many benchmarks and automatic metrics have been proposed, they differ substantially in what they define as a hallucination, what evidence they assume, and how well they agree with human judgment. This project will conduct a systematic survey of hallucination evaluation for large language models. We will develop a unified taxonomy, compare representative benchmarks and evaluation methods, analyze their reliability and practical trade-offs, and identify open problems for robust and reproducible evaluation.

1 Background and Motivation

Hallucination is a central obstacle to deploying large language models (LLMs) in knowledge-intensive and high-stakes settings. Existing work uses the term to describe several distinct failures, including contradiction with a source, unsupported claims, and conflict with established world knowledge. This lack of definitional agreement makes reported results difficult to compare (Venkit et al., 2024; Zhang et al., 2025). Widely used benchmarks such as TruthfulQA and HaluEval test different aspects of truthfulness and hallucination recognition (Lin et al., 2022; Li et al., 2023), while methods such as SelfCheckGPT, AlignScore, and FActScore rely on substantially different signals and units of analysis (Manakul et al., 2023; Zha et al., 2023; Min et al., 2023).

Recent reviews also show that human evaluation remains important but is often under-specified: annotation instructions, interfaces, agreement measures, and annotator information are not consistently reported (Schmidtova et al., 2025). A focused synthesis of evaluation methods is therefore needed to clarify what each method measures, which assumptions it makes, and when its scores can be trusted. Rather than surveying hallucination mitigation broadly, this project concentrates on the measurement problem.

2 Scope and Research Questions

The survey will cover methods and resources for evaluating hallucinations in text generated by LLMs. It will include open-ended question answering, summarization, dialogue, long-form generation, and retrieval-augmented generation when the central concern is factuality or faithfulness. Multimodal hallucination and mitigation-only studies will be outside the primary scope.

The review will address three research questions:

1. **RQ1: Definition and target.** How do studies define hallucination, and which evaluation unit, task, and error type do they target?
2. **RQ2: Methodological taxonomy.** What evidence, supervision, and computational mechanisms do current evaluation methods use?
3. **RQ3: Reliability and applicability.** How well do methods agree with human judgments and generalize across models, domains, languages, and generation tasks, and what are their cost and reproducibility trade-offs?

3 Methodology

We will follow a transparent literature-review protocol. Candidate studies will be collected from the ACL Anthology, Google Scholar, ACM Digital Library, IEEE Xplore, and arXiv using combinations of terms including *LLM*, *hallucination*, *factuality*, *faithfulness*, *evaluation*, *benchmark*, and *fact verification*. We will prioritize work from 2020 onward and use backward and forward citation tracing to identify influential earlier work and recent follow-ups. At least 20 representative academic papers will be included in the final report; we expect to screen a larger candidate set and record explicit inclusion and exclusion decisions.

For each included study, we will extract its hal-

lucination definition, task and domain, evaluation granularity, required evidence, supervision level, output type, datasets, human-evaluation protocol, reported correlation or agreement, model access requirements, cost, and released resources. The preliminary taxonomy will organize methods into five families: (1) human evaluation, (2) reference- or evidence-based metrics, (3) trained factuality or entailment evaluators, (4) self-consistency and uncertainty methods, and (5) LLM-as-a-judge methods. We will refine this taxonomy iteratively when studies do not fit the initial categories.

The comparison will emphasize validity rather than leaderboard scores alone. We will examine whether a metric measures factual correctness, source faithfulness, or merely response plausibility; whether its claims are supported by human agreement; and whether it remains stable under model, prompt, domain, or language changes. Benchmark contamination, judge-model bias, dependence on retrieved evidence, annotation inconsistency, and reproducibility will be treated as cross-cutting concerns.

4 Expected Outcomes

The project will produce: (1) a reproducible literature-selection record; (2) a unified taxonomy connecting hallucination definitions to evaluation methods; (3) comparison matrices of representative benchmarks and metrics; (4) a critical analysis of validity, robustness, cost, and applicability; and (5) recommendations for selecting evaluation methods in different NLP settings. The final report will also identify open directions, including fine-grained error localization, multilingual evaluation, reliable LLM-as-a-judge protocols, evaluation of long-form and retrieval-augmented outputs, and stronger meta-evaluation against carefully designed human studies.

5 Team Members and Responsibilities

The team will use the following division of labor. The team leader will participate in and take responsibility for every workstream, while each member also owns a clearly defined set of deliverables:

- **Zichen Cheng (Team Leader):** participates in and is responsible for all parts of the project, including scope definition, review protocol, literature search and screening, taxonomy design, evidence extraction, comparative analysis, report writing and integration, citation and quality control, slide preparation, and presentation coordination;
- **Yiming Fan:** leads the systematic literature search and screening log, analyzes hallucination definitions and error types, and prepares the conceptual foundation of the taxonomy;
- **Jin Li:** collects and analyzes benchmark datasets, reference- and evidence-based metrics, and trained factuality evaluators, and contributes the corresponding comparison evidence;
- **Sangduojie Yang:** analyzes self-consistency, uncertainty, and LLM-as-a-judge methods, with emphasis on robustness, generalization, and judge-model bias;
- **Lei Zhang:** analyzes human-evaluation protocols, reproducibility, computational cost, and open challenges, and leads the construction of comparison tables and presentation materials.

All five members will participate in paper screening, taxonomy refinement, critical discussion, writing, proofreading, and the final presentation. Zichen Cheng will review and coordinate every component to ensure consistency and overall quality. Individual screening and writing decisions will be recorded to make contributions clear.

6 Project Timeline

- **Stage 1—Scope and search protocol:** finalize definitions, databases, search strings, and selection criteria; build the candidate pool.
- **Stage 2—Screening and extraction:** assign papers across the five members, verify eligibility, and complete a shared evidence table.
- **Stage 3—Taxonomy and comparative analysis:** reconcile labels, refine the taxonomy, compare methods, and prepare the midterm results.
- **Stage 4—Synthesis and delivery:** write the ACL-format report, audit all citations and claims, prepare slides, and rehearse the presentation.

References

Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2023. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252.

Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100.

Patricia Schmidtova, Eduardo Calo, Simone Balloccu, Dimitra Gkatzia, Rudali Huidrom, Mateusz Lango, Fahime Same, Vilem Zouhar, Saad Mahamood, and Ondrej Dusek. 2025. Do my eyes deceive me? a survey of human evaluations of hallucinations in NLG. In *Proceedings of the 18th International Natural Language Generation Conference*, pages 60–79.

Pranav Narayanan Venkit, Tatiana Chakravorti, Vipul Gupta, Heidi Biggs, Mukund Srinath, Koustava Goswami, Sarah Rajtmajer, and Shomir Wilson. 2024. An audit on the perspectives and challenges of hallucinations in NLP. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6528–6548.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11328–11348.

Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, Longyue Wang, Anh Tuan Luu, Wei Bi, Freda Shi, and Shuming Shi. 2025. Siren's song in the AI ocean: A survey on hallucination in large language models. *Computational Linguistics*, 51(4).